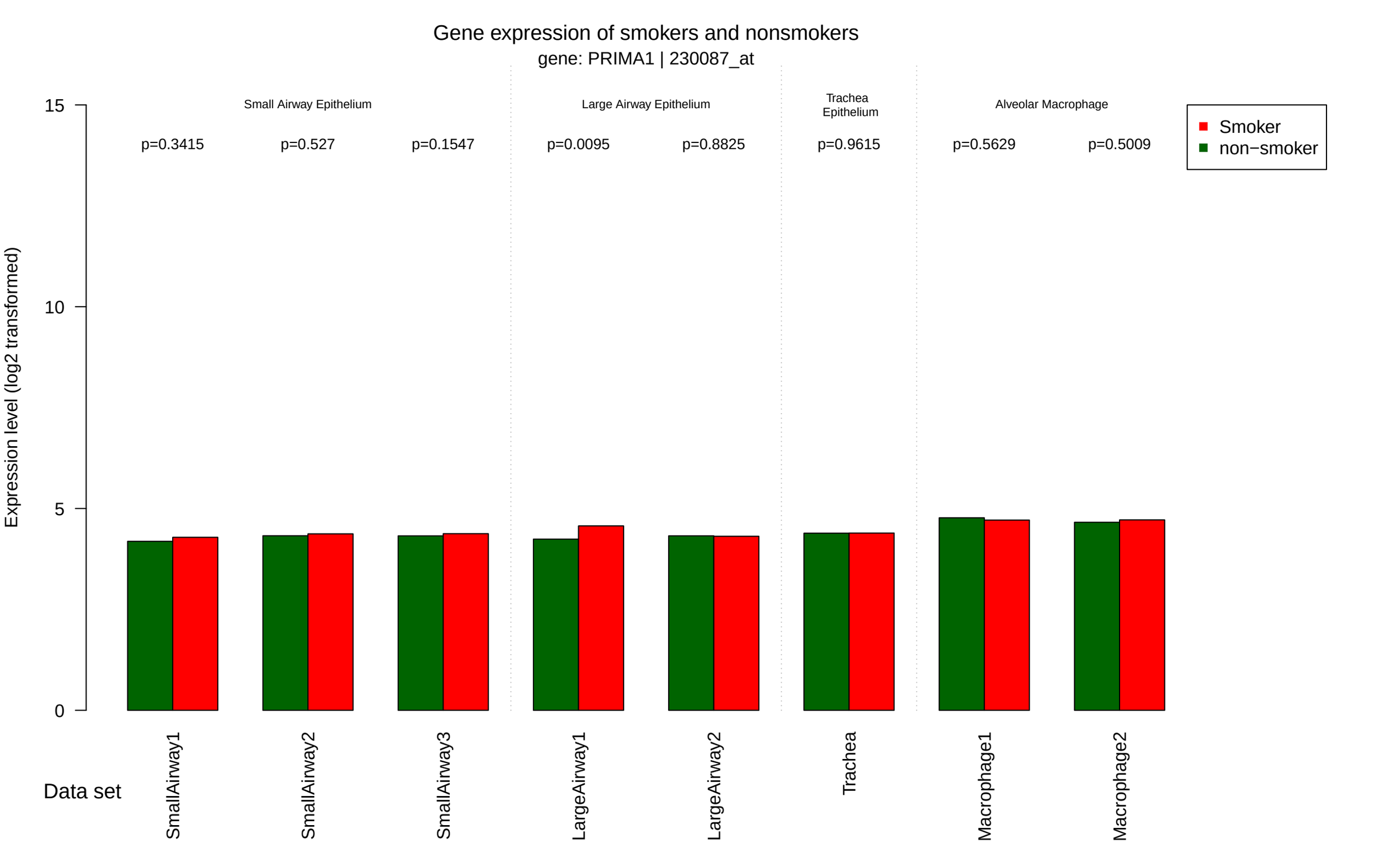
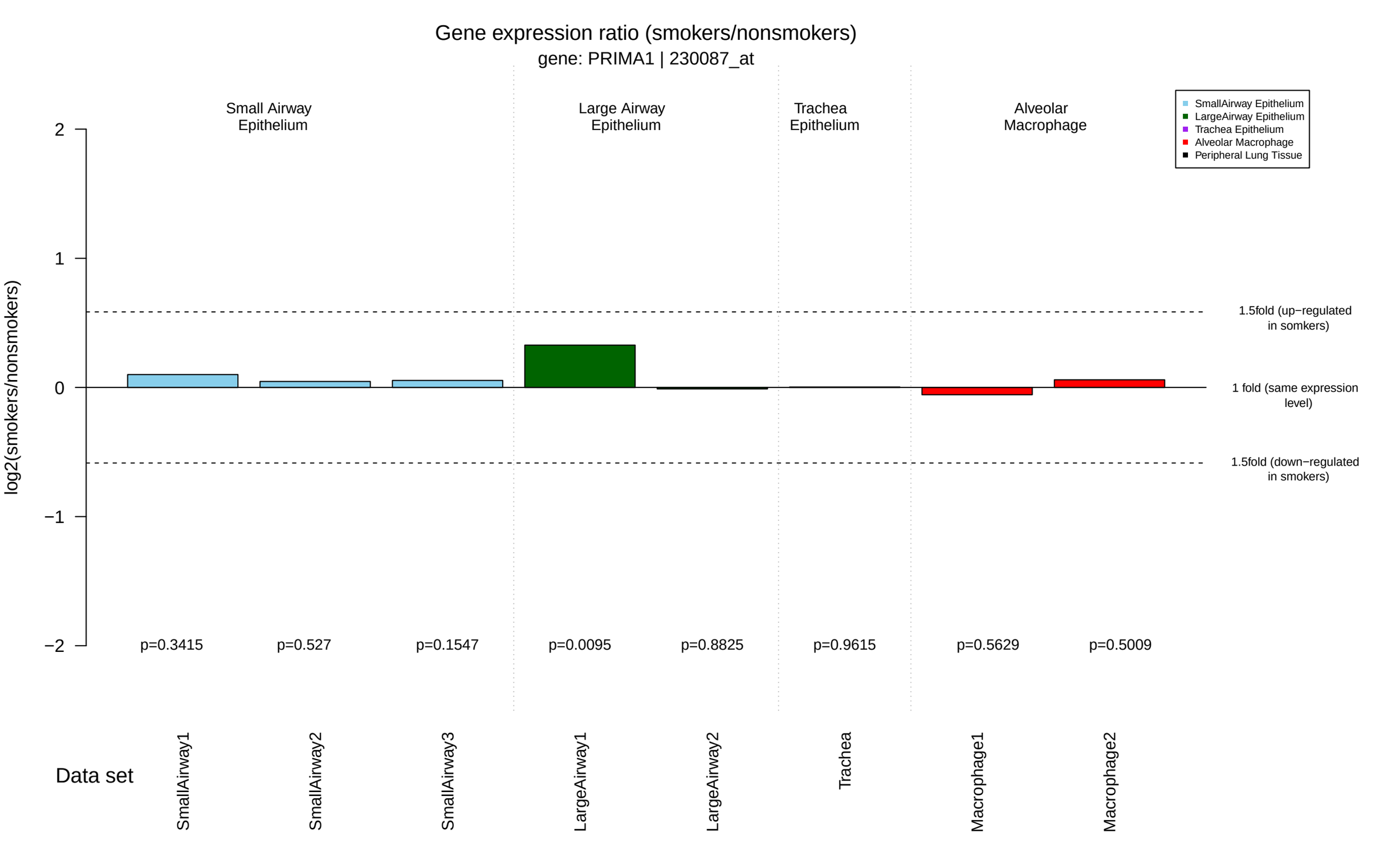
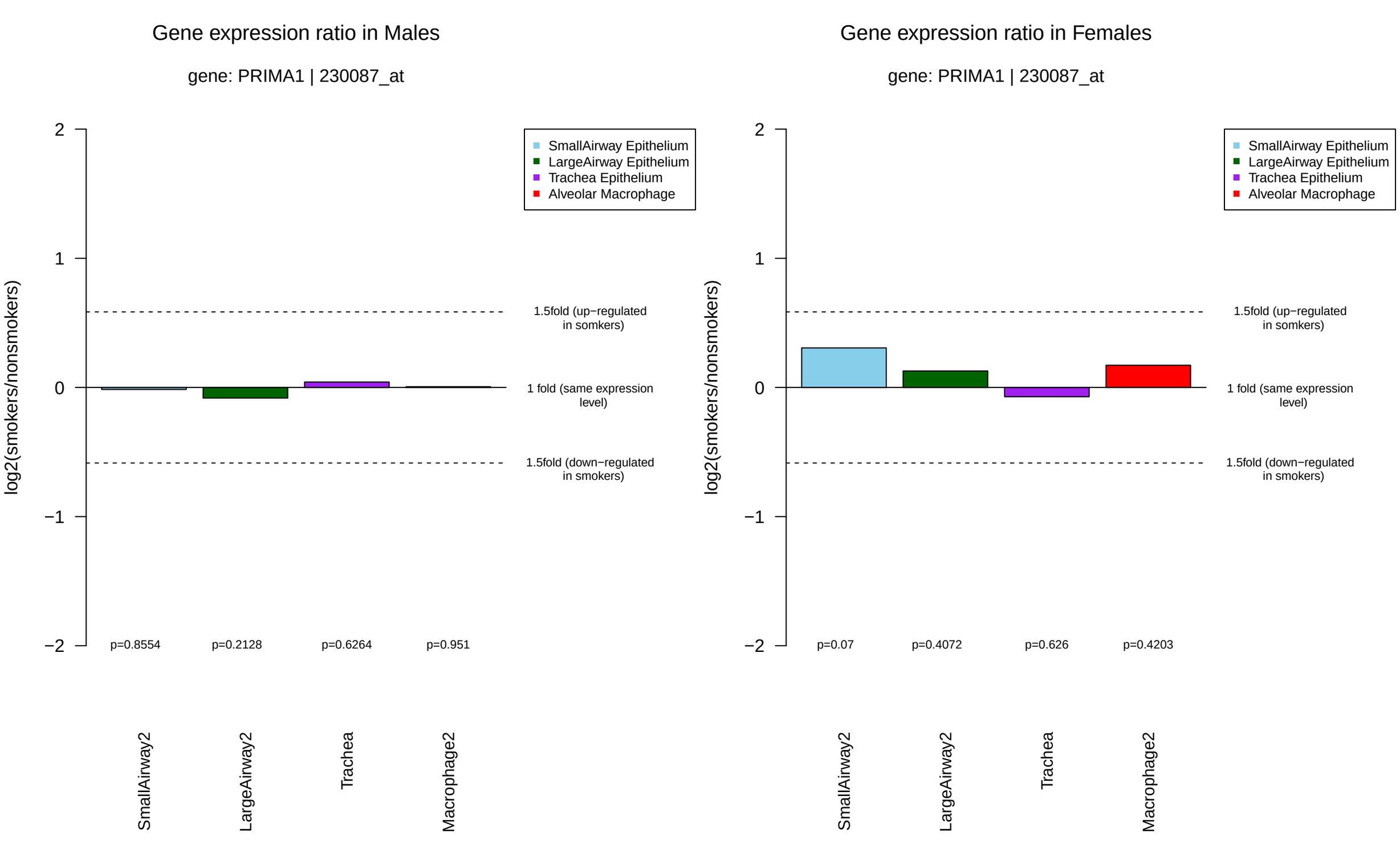
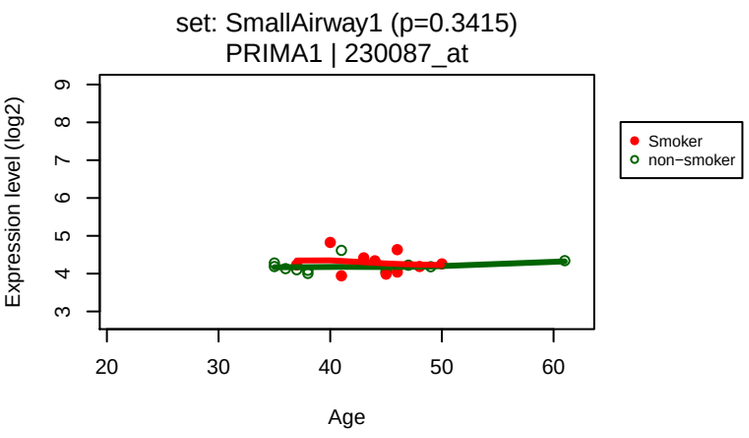
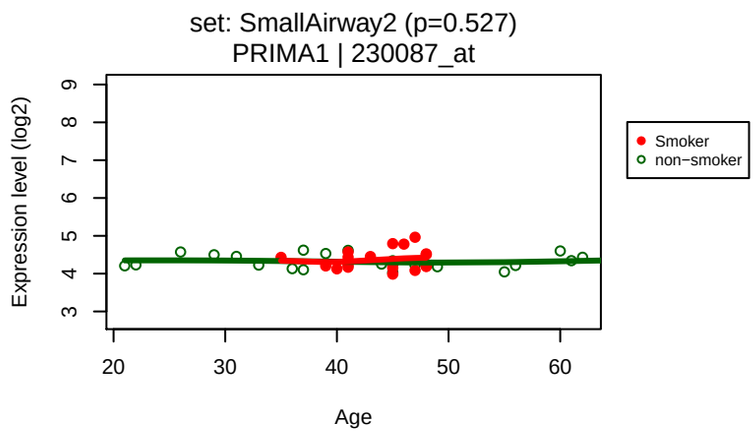
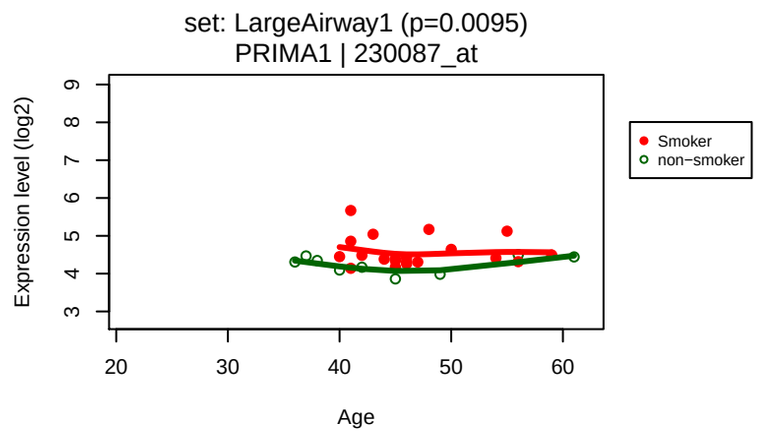

Gene expression of smokers and nonsmokers
gene: PRIMA1 | 230087_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3415
p=0.527
p=0.1547
p=0.0095
p=0.8825
p=0.9615
p=0.5629
p=0.5009
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

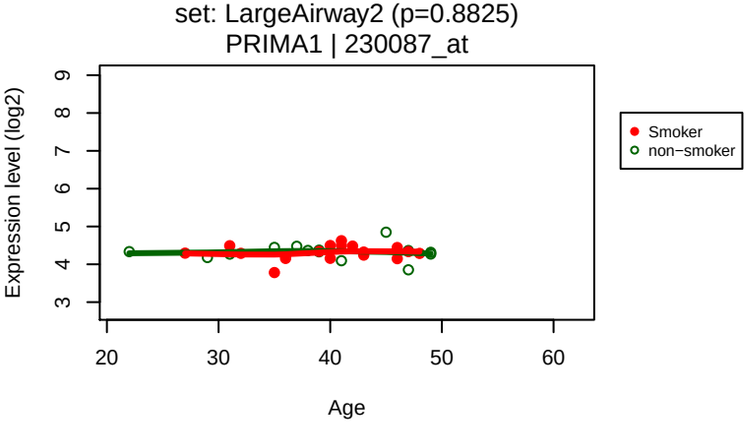

Gene expression ratio (smokers/nonsmokers)
gene: PRIMA1 | 230087_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3415
p=0.527
p=0.1547
p=0.0095
p=0.8825
p=0.9615
p=0.5629
p=0.5009
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

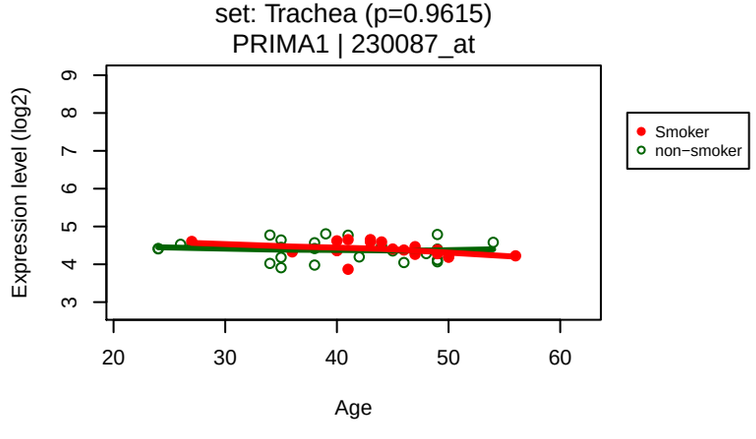

Gene expression ratio in Males
gene: PRIMA1 | 230087_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8554
p=0.2128
p=0.6264
p=0.951
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PRIMA1 | 230087_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.07
p=0.4072
p=0.626
p=0.4203
SmallAirway2
LargeAirway2
Trachea
Macrophage2

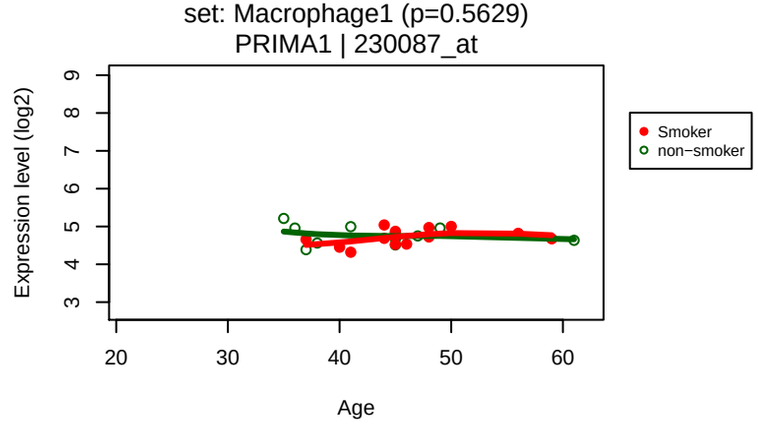

set: SmallAirway1 (p=0.3415)
PRIMA1 | 230087_at
Expression level (log2)
Age
Smoker
non−smoker

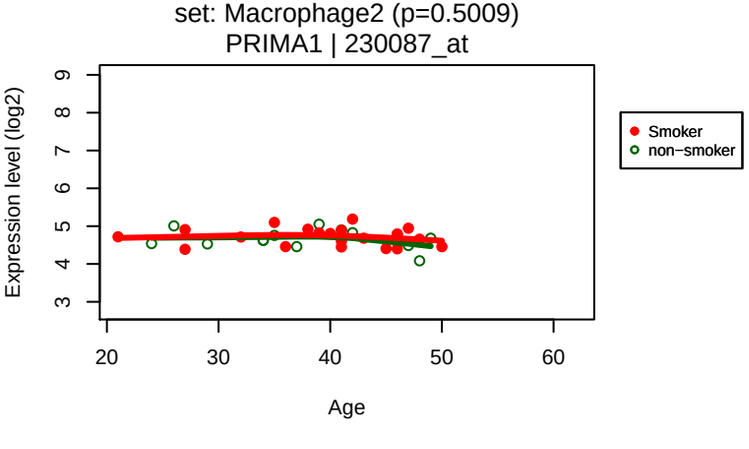

set: SmallAirway2 (p=0.527)
PRIMA1 | 230087_at
Expression level (log2)
Age
Smoker
non−smoker

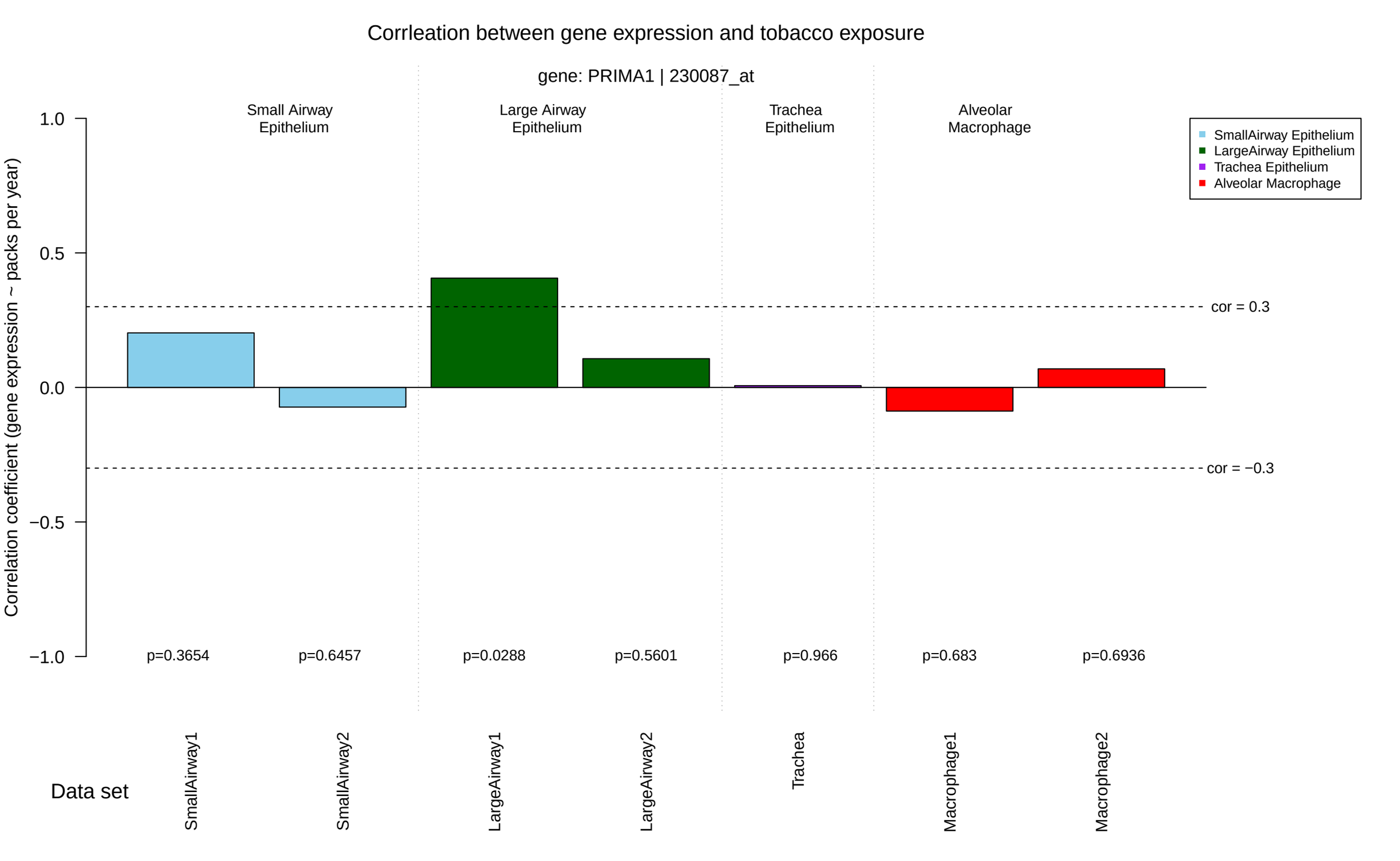

set: LargeAirway1 (p=0.0095)
PRIMA1 | 230087_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8825)
PRIMA1 | 230087_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9615)
PRIMA1 | 230087_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5629)
PRIMA1 | 230087_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5009)
PRIMA1 | 230087_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PRIMA1 | 230087_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3654
p=0.6457
p=0.0288
p=0.5601
p=0.966
p=0.683
p=0.6936
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2